

NIDHIL NAYUDU

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EDUCATION

University of Michigan

Jan. 2026 – Dec. 2026

Computer Science M.S.E; GPA: 4.00

Ann Arbor, MI

Relevant Coursework: Advanced Scalable Systems, Advanced Compilers, Computer & Network Security, Reinforcement Learning

University of Michigan: 7x Dean's List, 7x University Honors

Aug. 2022 – Dec. 2025

Computer Science B.S.E, Minor in Business; GPA: 3.85

Ann Arbor, MI

Relevant Coursework: Data Structures & Algorithms, Web Systems, Distributed Systems, Operating Systems, Computer Security, Networking, Machine Learning, Computer Science Theory

EXPERIENCE

Capital One

June 2026 - Aug. 2026

Software Engineer Intern

New York, NY

- Architected a server-side TypeScript BFF deployed across 2 AWS Lambdas to bypass $N - 3$ client version constraints, accelerating product velocity by cutting feature rollout time from over 2 months to under 5 minutes for 200,000 unique monthly users.
- Restructured CMS architecture by consolidating a 3D matrix (platform/language/account) into unified base documents with sparse deltas, cutting authoring surface area by $\sim 70\%$, reducing risk of drift, and dropping update effort from $O(P \times L \times A)$ to $O(1)$.
- Implemented in-memory OAuth2 token caching across warm Lambda execution environments, bypassing redundant Secrets Manager and token endpoint calls to cut amortized auth latency from $\sim 200ms$ to $< 1ms$ and prevent rate throttling.
- Leveraged custom Claude Code skills and agents to streamline development, creating a 4-agent PR review skill (Security, Testing, Architecture, Quality) that caught OWASP vulnerabilities and breaking changes pre-submission to speed up team code reviews.

CGI

June 2025 - Aug. 2025

Software Development Intern

New York, NY

- Incorporated Model Context Protocol (MCP) into our Agentic AI framework for SDLC automation, decoupling agent logic from client systems to enable server-side updates that cut client integration time by 42%.
- Configured GitLab CI/CD pipelines and core Agentic AI agents (BDD test generation/execution, Dev Environment, Pipeline Orchestration), automating ATDD validation and boosting test coverage by 8% across over 10 modules.
- Spearheaded architectural redesign over three Agile sprints, aligning client and executive goals by leading stakeholder meetings and creating system diagrams that halved design approval time and reduced misalignment.

University of Michigan EECS 280: Programming & Intro Data Structures

Aug. 2024 - Present

Teaching Assistant - Office Hours Lead

Ann Arbor, MI

- Lead lab sections and office hours to assist over 1,000 students, reinforcing core CS concepts and providing personalized support.
- Coordinate office hours operations for a team of teaching assistants by managing schedules, communicating updates, and organizing additional support during peak office hours with queues of up to 50 students around project deadlines.
- Facilitate understanding of concepts including recursion and polymorphism by refining lecture material and providing guidance.

PROJECTS

SafePrompt | 4th Place, Capital One Hackathon | TypeScript, Python, FastAPI, Chrome MV3, Presidio, DistilBERT, React

- Created a Chrome extension that intercepts DOM prompt submissions and image attachments across ChatGPT, Claude, and Gemini, enforcing strictly local, on-device sanitization so sensitive enterprise data never hits third-party LLMs.
- Designed the detection engine as a 2-stage pipeline that runs regex to catch obvious PII, and then routes complex text and images to a locally hosted FastAPI backend running Presidio, DistilBERT, and OCR with a strict block-on-timeout rule to prevent data leaks.
- Connected a React admin dashboard to push policy rules to the backend via periodic polling and monitor live, non-sensitive audit logs.

Prefetching Retrieval-Augmented Generation | Python, FastAPI, RAG, FAISS, NLP

- Engineered a Python/FastAPI speculative prefetching pipeline for interactive RAG, streaming user input as they type to execute asynchronous FAISS vector searches that cut mean latency by 18%.
- Developed a sliding-window extractor and draft-scoped session cache to mitigate intent drift, alongside a submit-time verification gate achieving a 99% pass rate.
- Evaluated tunable execution policies on a LoTTE-aligned benchmark, optimizing background scheduling to quantify a speed-cost tradeoff that reduced backend CPU load by 21.2%.

Thread Library | C++, Multithreading, Unix, OS, Unit Testing

- Crafted an advanced C++ thread library with meticulous implementation of Mutexes, Threads, and Conditional Variables, featuring essential functions like lock, unlock, signal, broadcast, wait, thread yield, and thread join.
- Implemented memory management mechanisms to prevent leaks and enhance library reliability in concurrent environments.
- Enforced comprehensive unit tests to assure the library's soundness, identify and rectify errors, and guarantee optimal performance in diverse scenarios.

ask485?: Search Engine | Python, React.js, Hadoop, Distributed Systems, Flask, Multithreading, APIs

- Created a Google-style scalable search engine with a React.js frontend and Python backend to return ranked web search results.
- Constructed a segmented inverted index using a MapReduce pipeline with Hadoop streaming for efficient query retrieval.
- Employed Flask Python framework to develop REST APIs and route requests with backend index and search servers.

TECHNICAL SKILLS

Languages : Python, C/C++, TypeScript, JavaScript, Go, Bash Scripting, HTML/CSS, SQL

Technologies : React, Next.js, Node.js, Git, Claude Code, AWS, Tensorflow, PyTorch, Docker, Linux

Concepts : Object-Oriented Programming, Machine Learning, Multithreading, Networks, REST APIs, Agile Dev, CI/CD